

10(a)	<i>eight from:</i> 1 chiasmata formation ; 2 crossing over ; 3 exchange of genetic material ; 4 between non-sister chromatids of homologous chromosomes ; 5 at prophase 1 ; 6 new combination of alleles ; 7 linkage groups broken ; 8 independent / random, assortment, of homologous pairs / bivalents / AW ; 9 in metaphase 1 ; 10 2^n combinations / very large number of combinations ; 11 independent / random, assortment of, chromatids / chromosomes (correct context) ; 12 in metaphase 2 ; 13 <i>ref. to</i> mutation ;	8
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Question	Answer	Marks
10(b)	<i>seven from:</i> 1 <i>ref. to TYR</i> gene ; 2 normal gene product is tyrosinase ; 3 tyrosine converted to, DOPA / dopaquinone ; ora 4 melanin / pigment, made ; ora 5 in melanocytes ; 6 mutant allele is recessive ; 7 tyrosinase, not produced / inactive ; 8 affects, hair / skin / irises ; 9 only in homozygous recessive people ;	7